

KubernetesError 3: Pending Pods

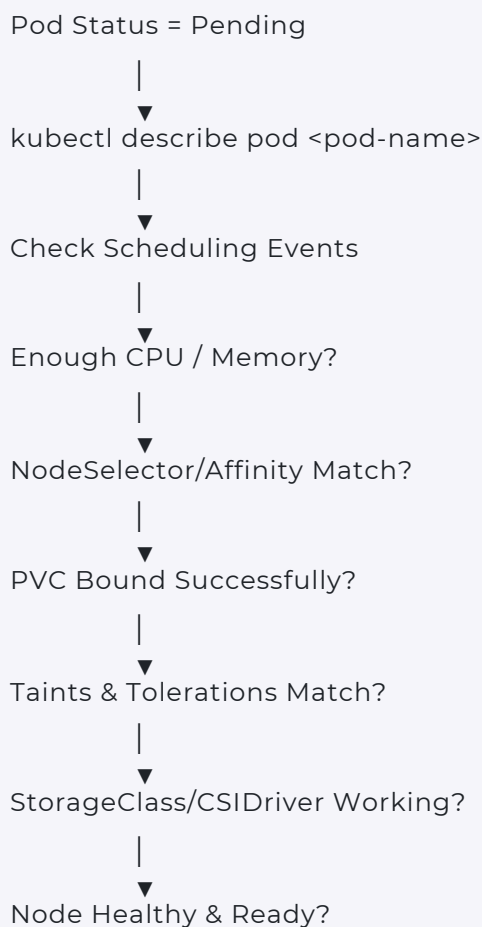
- A Pod remains in the Pending state because Kubernetes is unable to schedule it on any node.

Common Reasons

-  Insufficient CPU or memory on the cluster
-  No available nodes
-  Unbound PersistentVolumeClaim (PVC)
-  Node Selector or Node Affinity mismatch
-  Taints and Tolerations mismatch
-  Image pull secrets or storage provisioning issues



Pending Pod Troubleshooting Flow





Kubernetes Pending Pods — Common Reasons & Solutions

 Cause	 Description	 Solution	 Useful Commands
 Pod Scheduling Failure	Scheduler cannot place the Pod on any node	Check Pod events to identify the scheduling reason	<pre>kubectl describe pod <pod-name></pre>
 Insufficient CPU or Memory	Nodes don't have enough available resources	Verify node capacity and resource usage; add nodes or free resources	<pre>kubectl get nodes kubectl describe node <node-name></pre>
 Resource Requests Exceed Capacity	CPU/Memory requests are larger than available resources	Reduce resource requests/limits or scale the cluster	<pre>kubectl edit deployment <deployment-name></pre>
 Node Selector / Node Affinity Mismatch	No node matches the Pod's scheduling rules	Update or remove <code>nodeSelector</code> or affinity rules	<pre>kubectl describe pod <pod-name></pre>
 PersistentVolumeClaim (PVC) Pending	Required storage is not available	Verify PVC, StorageClass, PV, and CSI Driver	<pre>kubectl get pvc kubectl describe pvc <pvc-name></pre>
 Taints and Tolerations Mismatch	Node taints prevent Pod scheduling	Add matching tolerations or use another node	<pre>kubectl describe node <node-name></pre>
 Unschedulable Node	Node is cordoned or marked unschedulable	Uncordon the node or schedule on another node	<pre>kubectl get nodes</pre>
 Cluster Autoscaler Delay	New nodes have not been provisioned yet	Wait for autoscaling or add nodes manually	<pre>kubectl get nodes</pre>

 Cause	 Description	 Solution	 Useful Commands
 CSI Driver / Storage Issues	Storage provisioner cannot create a volume	Check CSI driver status and StorageClass	kubectl get storageclass kubectl get csidrivers
 Resource Quota / LimitRange	Namespace quota or limits prevent scheduling	Check namespace quotas and adjust limits	kubectl describe resourcequota

Useful Troubleshooting Commands

 Command	 Purpose
kubectl get pods	Check Pod status
kubectl describe pod <pod-name>	View scheduling events
kubectl get nodes	Check node status
kubectl describe node <node-name>	Inspect node resources and taints
kubectl top nodes	Check CPU and memory utilization
kubectl get pvc	Verify PVC status
kubectl describe pvc <pvc-name>	Check why a PVC is pending
kubectl get storageclass	Verify StorageClass configuration
kubectl get events --sort-by=.lastTimestamp	Review recent cluster events

Interview Answer

Q: How do you troubleshoot a Pod stuck in the Pending state ?

- Check the Pod events using **kubectl describe pod** .
- Verify node CPU and memory availability.
- Review resource requests and limits.
- Check **nodeSelector** and node affinity rules.
- Verify that the PVC is bound and the StorageClass/CSI driver are working.
- Inspect node taints and Pod tolerations.

- Check node health and namespace quotas if necessary.

Interview One-Liner

- A Pod remains in the Pending state when Kubernetes cannot schedule it onto a node.
- The most common causes are **insufficient CPU** or **memory** , **excessive resource requests** , **nodeselector** or **affinity mismatches** , **pending PVCs** , **taint and toleration mismatches** , **storageprovisioningissues** , or **unschedulablenodes** .
- The first step is always to inspect the scheduling events with **kubect! describe pod** 